Ignacio Contreras

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Resarch Interests

I am interested in the intersection of convex optimization and analysis with applications to inverse problems in medical Imaging. Other topics include computational harmonic analysis, foundations of data science, compressed sensing, quantum information theory.

Current Position

Research Assistant *Santiago, Chile*
Biomedical Imaging Center, Pontificia Universidad Católica de Chile (PUC) *Aug 2025 – Now*

Education

Pontificia Universidad Católica de Chile (PUC) *Aug 2022- Dec 2024)*
M.Sc. in Engineering, advised by Carlos Sing Long

- Thesis: *Lifting to Measure Spaces Techniques for Nonlinear Inverse Problems.* Available here

Pontificia Universidad Católica de Chile (PUC) *2021 – 2022*
Mathematical and Computational Engineering (Professional Degree)

- Coursework (mostly graduate courses): Functional Analysis, PDEs, Advanced Optimization, Applications of Functional Analysis and PDEs to Engineering, Probability Theory, Topics in Inverse Problems, High-Dimensional Probability, Regression Analysis, Parallel Algorithms in Scientific Computing, Biomedical Image Formation, Mathematical Foundations of Machine Learning, etc.

Pontificia Universidad Católica de Chile (PUC) *2017 – 2021*
B.Sc. (Licenciatura) in Engineering

- **Major in Engineering Physics**
- Coursework: Quantum Mechanics, Electromagnetic Theory, Statistical Mechanics, Mathematics Methods, Waves and Optics, Classical Mechanics, etc.
- **Minor in Applied Mathematics**
- Coursework: Real Analysis, Measure Theory, Fourier Analysis, Scientific Computing, Statistical Inference, Discrete Mathematics, Optimization.

Papers & Preprints

1. **I. Contreras-Zúñiga**, M. Lambert, B. Palacios, C. Tejos, C. Milovic. Dipole-lets: A new multiscale decomposition for MR Phase and Quantitative Susceptibility Mapping. arXiv preprint arXiv:2510.24705

Conference Abstracts

1. **I. Contreras-Zúñiga**, C. Tejos, C. Milovic. Demixing Streaking for Quantitative Susceptibility Mapping. 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)
2. **I. Contreras-Zúñiga**, C. Tejos, C. Milovic. Detecting and Correcting Phase Unwrapping Artifacts for Quantitative Susceptibility Mapping. International Society for Magnetic Resonance in Medicine 34th Scientific Meeting, 2026
3. C. Milovic, **I. Contreras-Zúñiga**, M. Lambert, C. Tejos. Dipole-lets ($\mathfrak{D}$): a new multiscale framework for MR phase/QSM analysis and reconstruction. International Society for Magnetic Resonance in Medicine 34th Scientific Meeting, 2026

Talks & Posters

1. How can we reduce artifacts in Quantitative Susceptibility Mapping MRI? (poster) Inverse Problems in the Physical Sciences IPPhys2026 Summer School. Puerto Varas, Chile, Webpage here

2. Recovery of Atomic Measures Supported on the Complex Plane Through Convex Optimization (poster) Encuentro Nacional de Ingeniería Matemática 2024 ENIM 2024. Universidad de Chile. Webpage here

Research Experience

Undergraduate Research, advised by Carlos Sing Long — *Santiago, Chile*
Institute for Mathematical and Computational Engineering, PUC — *April 2021 – Sept 2021*

- Studied the atomic norm minimization for super-resolution of point sources and implemented an SDP approach in Julia using `Convex.jl`.

Summer Research in Mathematics, advised by Carlos Pérez-Arancibia — *Santiago, Chile*
Faculty of Mathematics, PUC — *Nov 2020 – Jan 2021*

- Studied Boundary Integral Equations (BIE) for 2D scattering for the Helmholtz equation and implemented in Julia an adjoint-based method for shape optimization with BIE constraints.

Undergraduate Research, advised by Clémentine Béchet — *Santiago, Chile*
Institute for Mathematical and Computational Engineering, PUC — *Aug 2020 – Dec 2020*

- Studied statistical model selection and implemented in OpenCV different distortion models for close-range photogrammetry for the MOONS project of the European Southern Observatory.

Summer Research in Physics, advised by Rafael Benguria — *Santiago, Chile*
Faculty of Physics, PUC — *Jan 2020*

- Studied three classic inverse problems in mathematical & physics. Inversion of the Radon Transform, "Can one hear the shape of a drum?" problem (non-isometric shape can have the same Laplacian eigenvalues) and electrical impedance tomography.

Other Relevant Course Projects — *Santiago, Chile*
Institute for Mathematical and Computational Engineering, PUC — *2020-2021*

- Topics in Inverse Problems: Studied Robust Principal Component Analysis (recovery of a matrix with a sparse component and a low-rank component) through the minimization of the ℓ^1 and nuclear norm with an implementation of an ADMM algorithm with application to separation of components on a surveillance camera.
- Fourier Analysis: Studied the ℓ^1 reconstruction and the recovery guarantees of compressed sensing with an implementation in Julia

Work Experience

Data Science Final Project Mentor — *Santiago, Chile*
Institute for Mathematical and Computational Engineering, PUC — *Aug 2024 – Jan 2026*

- Mentor for the final project (Actividad Final de Graduación) of the Master of Data Science PUC (Coursera). Mentored 3+ groups with weekly feedback on projects.
- TA on Intro to Data Science (Oct-Dic 2025)

Intern — *Antofagasta, Chile*
European Southern Observatory (ESO). Paranal Science Operations. — *Jan 2022 – Mar 2022*

- Professional Internship at Paranal Observatory, ESO (remote due to Covid Pandemic)
- Studied and implemented the detrended fluctuation analysis and other statistical methods for cloud and precipitation detection from the observatory's time series weather data in Python.

Teaching (class recitations)

Teaching Assistant († denotes graduate courses)
Institute for Mathematical and Computational Engineering, PUC

- Applications of PDEs and Functional Analysis in Engineering† — *2023-1st, 2025-1st*
- Advanced Optimization† — *2023-1st*
- Topics in Inverse Problems† — *2022-1st, 2023-1st*
- Fourier Analysis — *2021-2nd*

Teaching Assistant *Faculty of Mathematics, PUC*

- Topics in Microlocal Analysis and Inverse Problems† (scribe) *2023-2nd*
- Scientific Computing I *2020-1st, 2022-2nd*
- Calculus III *2021-1st*

Teaching Assistant
Faculty of Engineering, PUC

- Biomedical Imaging *2022-2nd*
- Electricity & Magnetism Lab *2021-Summer*
- Electricity & Magnetism (grader) *2020-1st*

Grants & Awards

HackSciML - Hackathon on Scientific Machine Learning *Santiago, Chile*
National Center for Artificial Intelligence (CENIA) *Oct 2023*

- Our team (3 people) got 1*st* place at the HackSciML organized by the National Center for Artificial Intelligence (graduate challenge). HackSciML webpage & Picture of the award (Instagram)

SIAM Student Chapter Certificate of Recognition. Press release, in spanish *April 2023*
Society for Industrial and Applied Mathematics (SIAM)

- For exceptional service to the SIAM-PUC Student Chapter.

IMC-SIAM Travel Award
Institute for Mathematical and Computational Engineering, PUC *July 2022*

- Travel award to represent the SIAM-PUC Student Chapter at the SIAM Annual Meeting 2022, Pittsburgh, PA.

Skills

Programming: Python, Julia, MATLAB, R, LaTeX

Languages: Spanish (native), English (advanced), French (beginning)

Activities

Organizer, SIAM-PUC Summer School. 200 years of Fourier Analysis. Webpage here *Jan 2023*

Organizer, Mathematical Engineering National Meeting ENIM 2022. Webpage here *Aug 2022*

President, SIAM-PUC Student Chapter *Aug 2021-Aug 2023*

Student Representative, Institute for Mathematical and Computational Engineering *April 2021-April 2024*

Vice President, SIAM-PUC Student Chapter *Aug 2020-Aug 2021*

Student Representative, Engineering Physics Major *April 2020-Aug 2021*